Automated Model Validation for Heat Pumps

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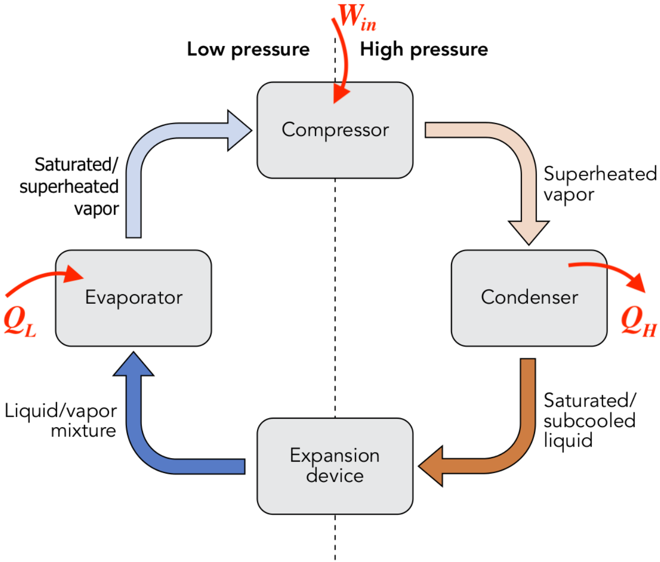
Presentation Overview

1. Introduction

- BDR Thermea: creating smart thermal comfort solutions to reduce the carbon footprint
- 1 Solution: Heat Pumps

2. Heat Pumps



W_{in}
Low pressure
High pressure
Compressor
Saturated/
superheated
vapor
Superheated
vapor
Evaporator
Q_L
Condenser
Q_H
Liquid/vapor
mixture
Saturated/
subcooled
liquid
Expansion
device

3. Project Objectives

- Create a pipeline that can automatically call a software on a model
- Expand the pipeline to more complex tasks, like automatic FMU generation
- Implement a method to automate model validation

4. Technical Components

- YAML - For pipeline configuration and scripting
- FMU/FMI - Functional Mock-up Unit/Interface for model exchange and simulation
- Dymola - Dynamic modeling laboratory for simulation and model development
- Thermal Systems

5. Achieved Goal: Repository Analysis

- Utilized Azure DevOps pipelines and .yml scripts for repository analysis.
- Tasks included:
  - Listing and counting .mo files.
  - Identifying files with specific strings for precise model extraction.
  - Locating key directories.
- Employed PowerShell and command line scripts for effective data handling.

6. Failed Task: Executing Dymola Commands in Pipeline

Issue: Encountered difficulties integrating Dymola commands into the Azure DevOps pipeline.

Intended Tasks with Dymola Commands:

1. **Exporting Modelica Models to FMU**
2. **Automation of Model Conversion Process**

Possible Cause: The issue is likely due to the Dymola license not being verified within the pipeline environment.

7. Post-Resolution Steps

- **Model Conversion Using Dymola:** Use `translateModelFMU` command for '.mo' to '.fmu' file conversion.
- **Model Validation with Empirical Data:**
  1. Acquire empirical data from a heat pump for benchmarking.
  2. Simulate and compare with real-world conditions.
  3. Calculate error margin between simulation results and empirical data.
- **Extending Validation to Other Models:** Apply the validation method to different models within the heat pump system.

8. Possible Advancements After Project Completion

- **Parameter Optimization:** Fine-tune model parameters for increased accuracy.
- **Scalability Testing:** Assess models in larger scales or different heat pump systems to check robustness.
- **Uncertainty Analysis:** Conduct analysis to quantify confidence in predictions and impact of parameters.
- **Predictive Maintenance Modeling:** Develop methods for anticipating maintenance needs, improving reliability.

9. Conclusion

- **Achievements:**
    - Gained in-depth understanding of heat pumps.
    - Utilized Azure DevOps for repository analysis.
- **Challenges:**
    - Faced major setbacks with Dymola integration into the pipeline.
    - Delayed access to Azure DevOps repositories.
- **Next objective:** Resolution of the Dymola issue should be the top priority for any future advancements in the project.

10. References

- BDR Thermea Group. (2023). About Us. Available at: https://www.bdrthermeagroup.com/en/about-us/
- Wikipedia contributors. Heat Pump. Available at: https://en.wikipedia.org/wiki/Heat_pump